

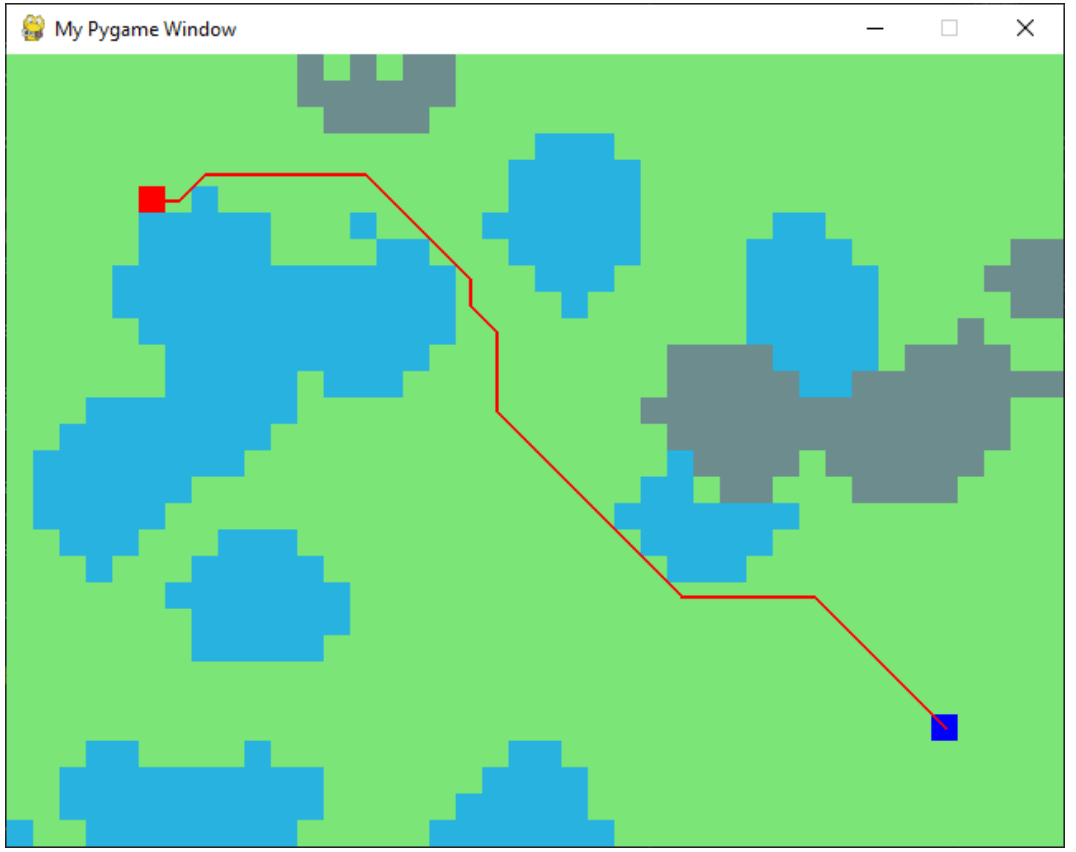
AI-programming, lecture 9
Miniproject 2 kick-off

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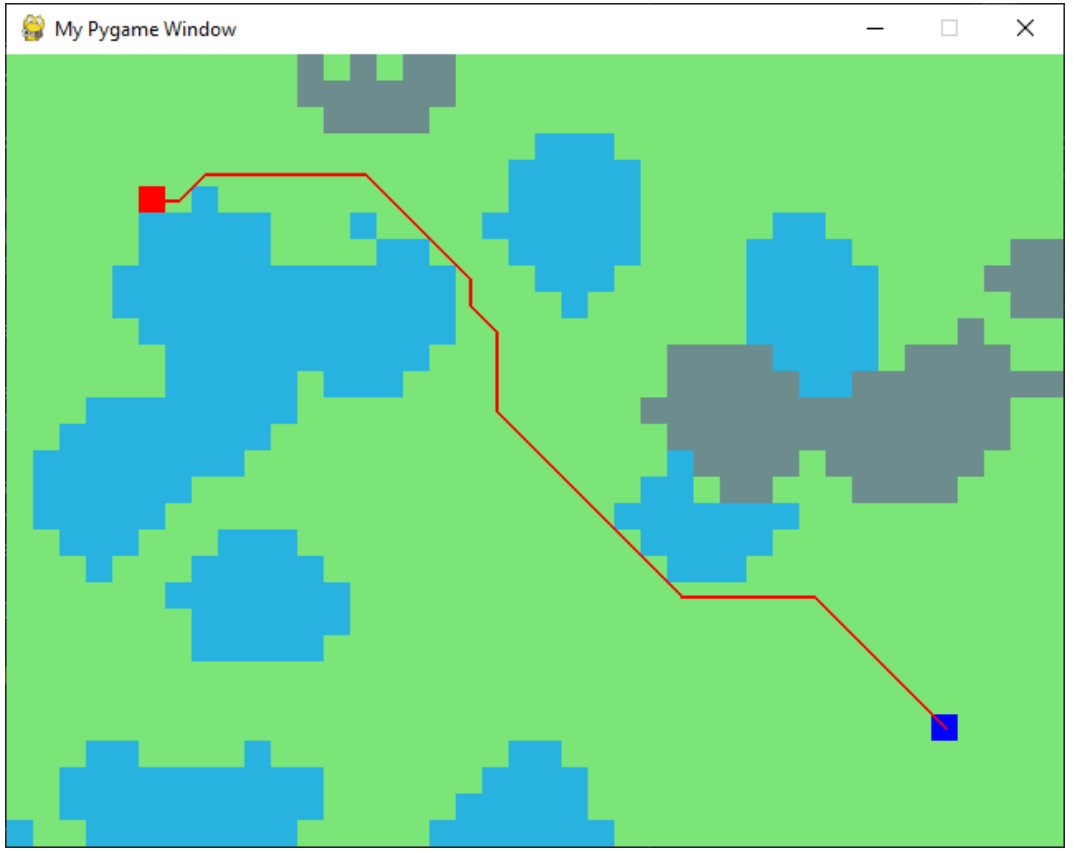
Miniprojekt 2

- Lav et program, der kan finde den optimale rute mellem to punkter i et landskab

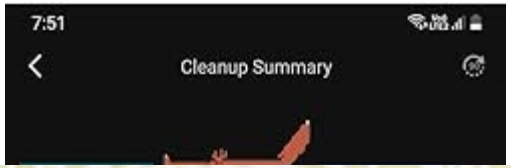


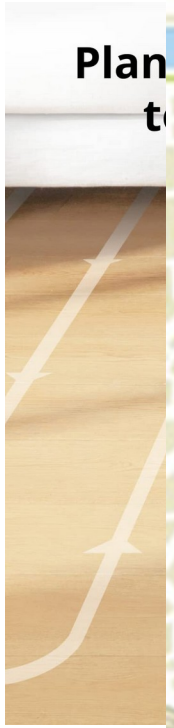
Miniprojekt 2

- Et program, der med en grafbaseret algoritme kan finde vej mellem to punkter.
- Et program, der som minimum indeholder et hardcoded kort, der skal findes vej på.
- Stien må ikke være hardcoded.
- Kortet og stien skal visualiseres på skærmen.
- I skal bruge funktioner, klasser og loops.



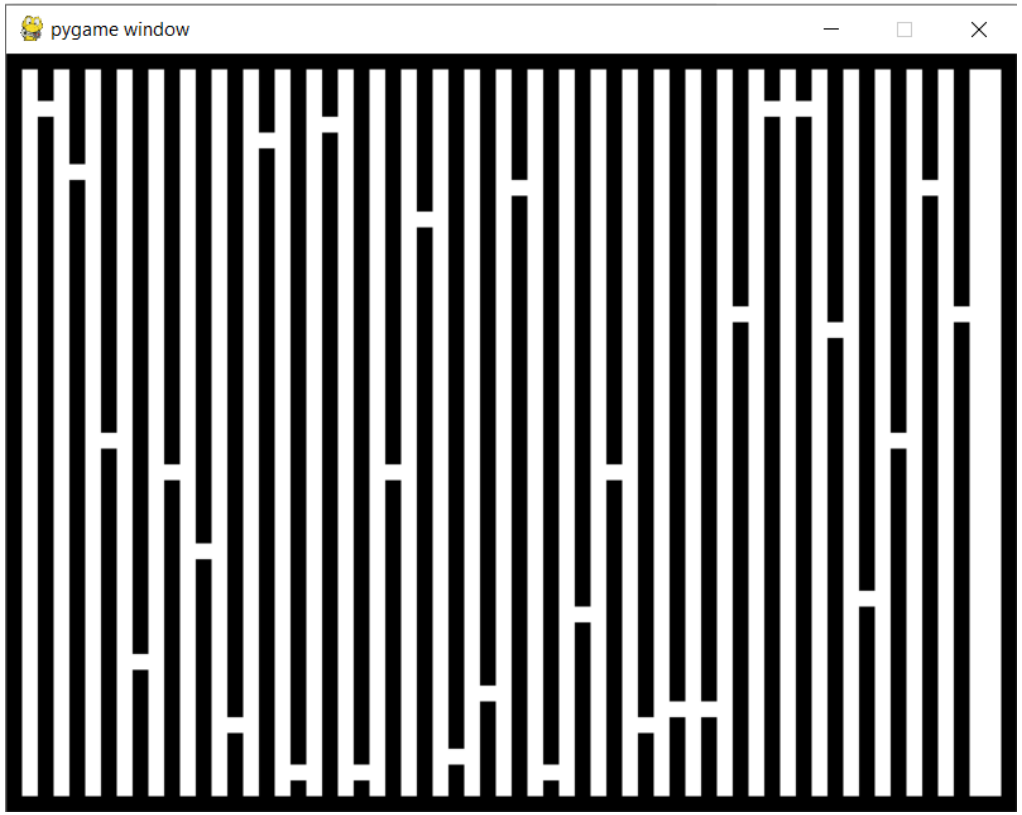
Miniprojekt 2

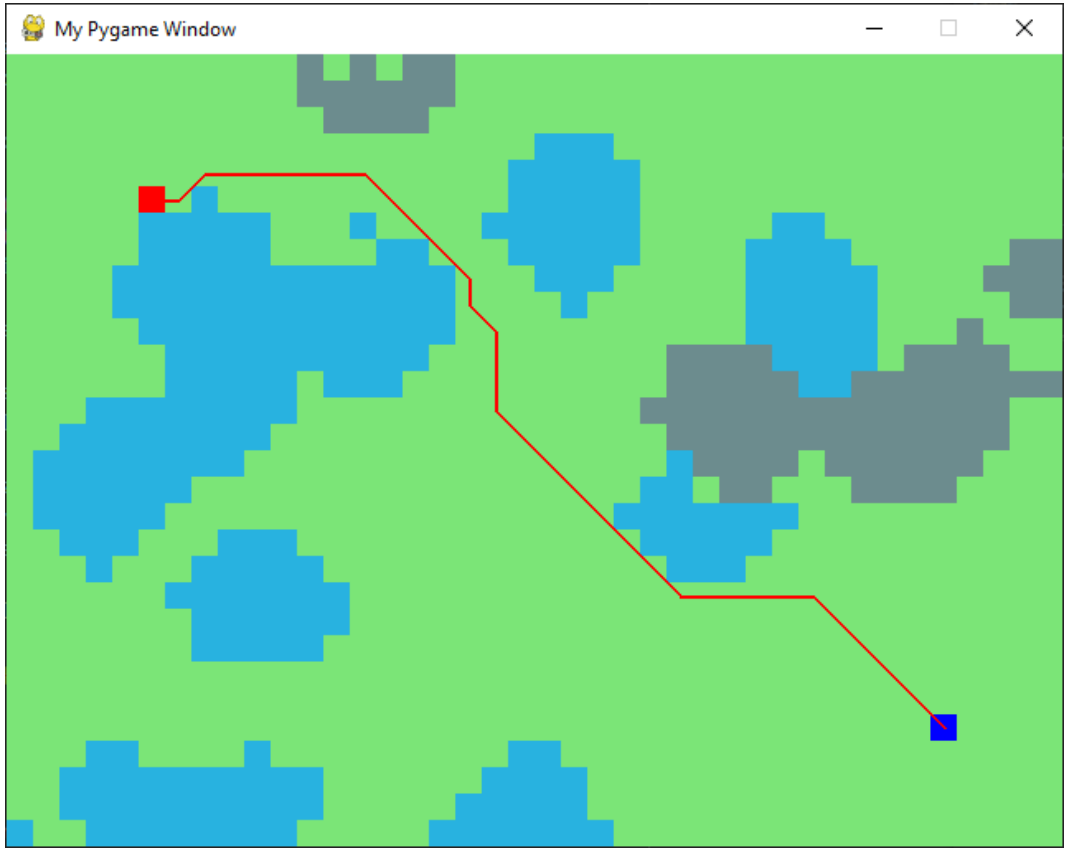






Miniprojekt 2





Your bible for this project!

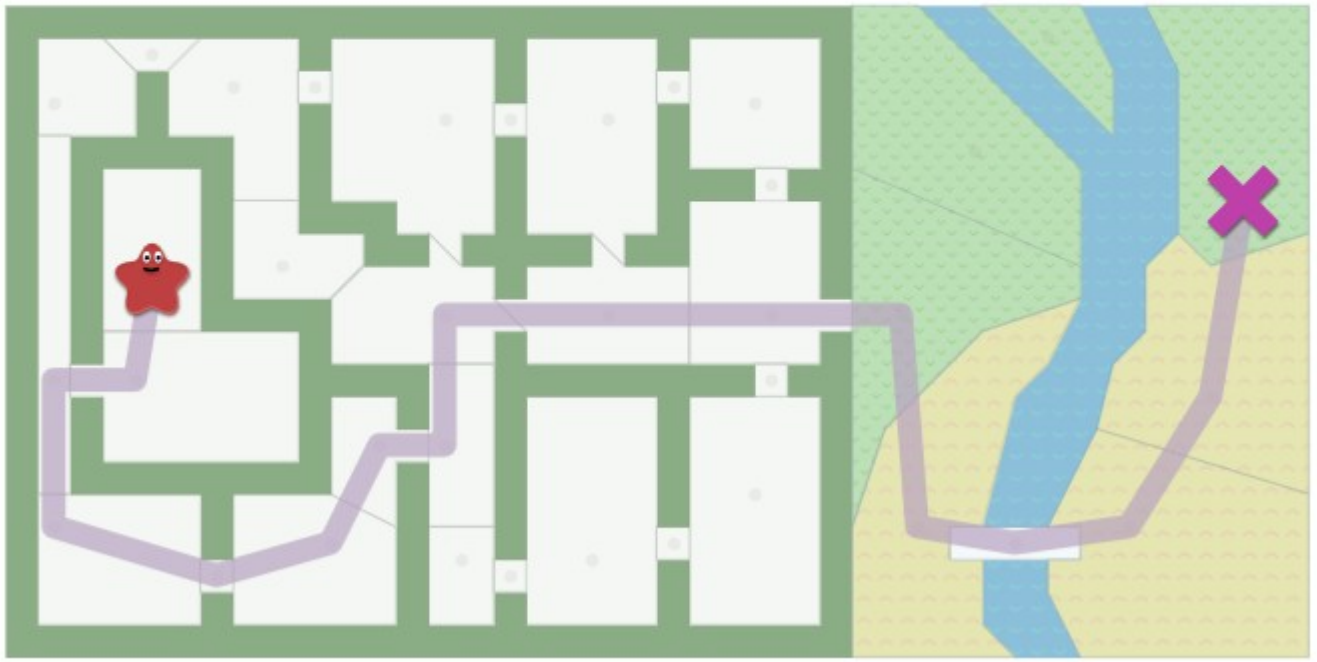
https://www.redblobgames.com/pathfinding/a-star/introduction.html



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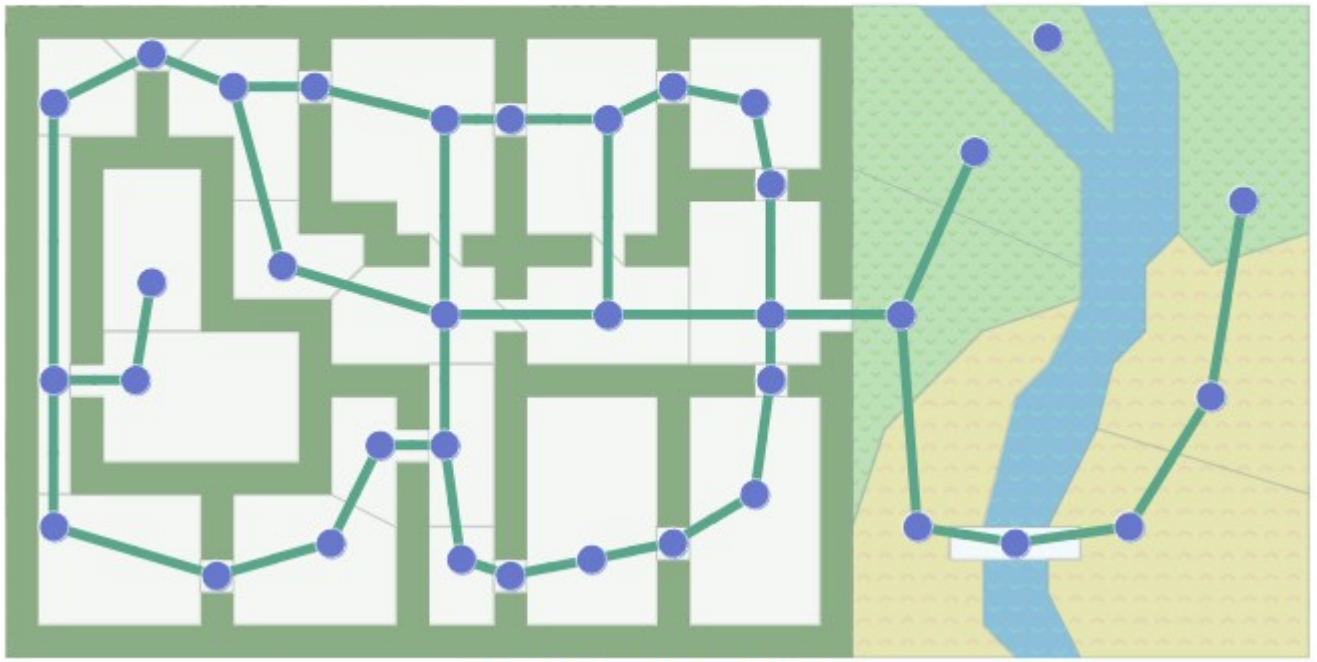
Graph-based pathfinding

Graph search algorithms let us find the shortest path on a map represented as a graph. Move the blob (start point) and cross (end point) to see the shortest path found by the **A* Algorithm**:



Graph-based pathfinding

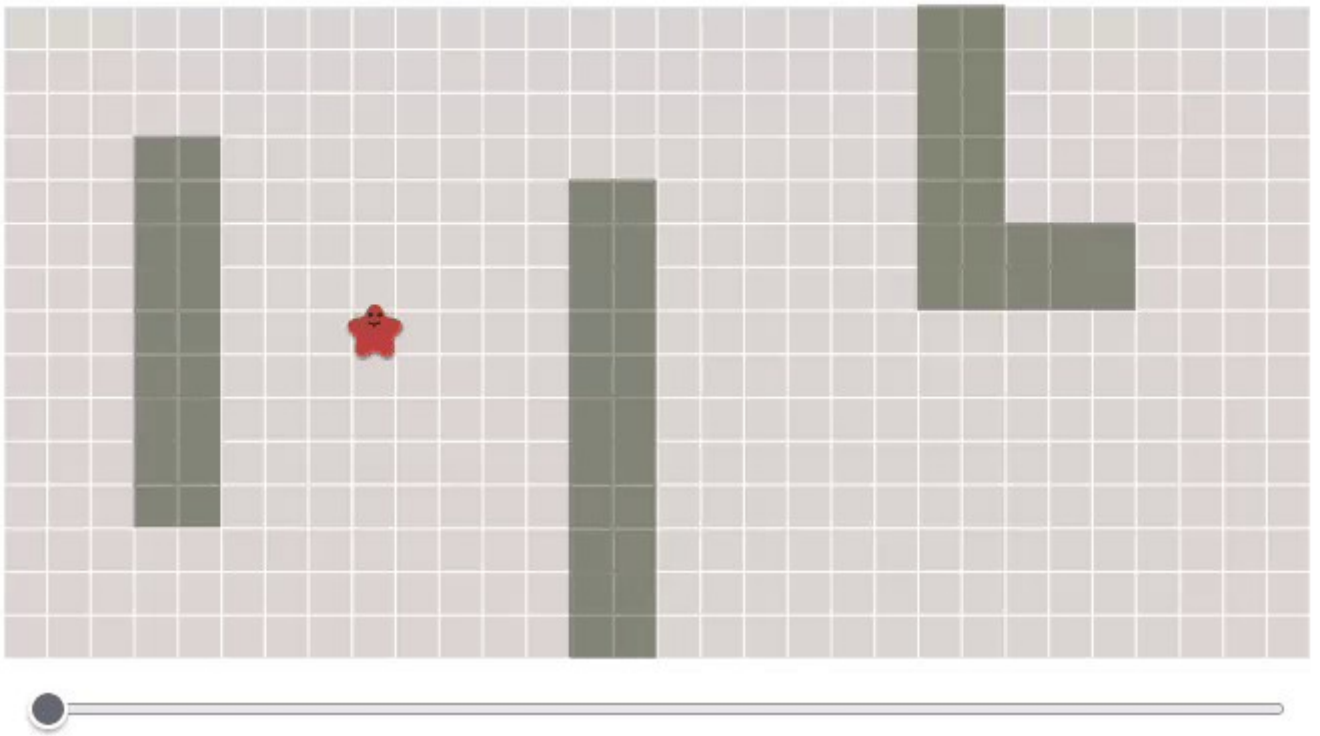
Input: Graph search algorithms, including A*, take a "graph" as input. A graph is a set of *locations* ● ("nodes") and the *connections* — ("edges") between them. Here's the graph I gave to A*:



Sprites by StarRaven - see footer for link

Grid search (simpler graph)

The key idea for all of these algorithms is that we keep track of an expanding ring called the ***frontier***. On a grid, this process is sometimes called "flood fill", but the same technique also works for non-grids. **Start the animation** to see how the frontier expands 



The code

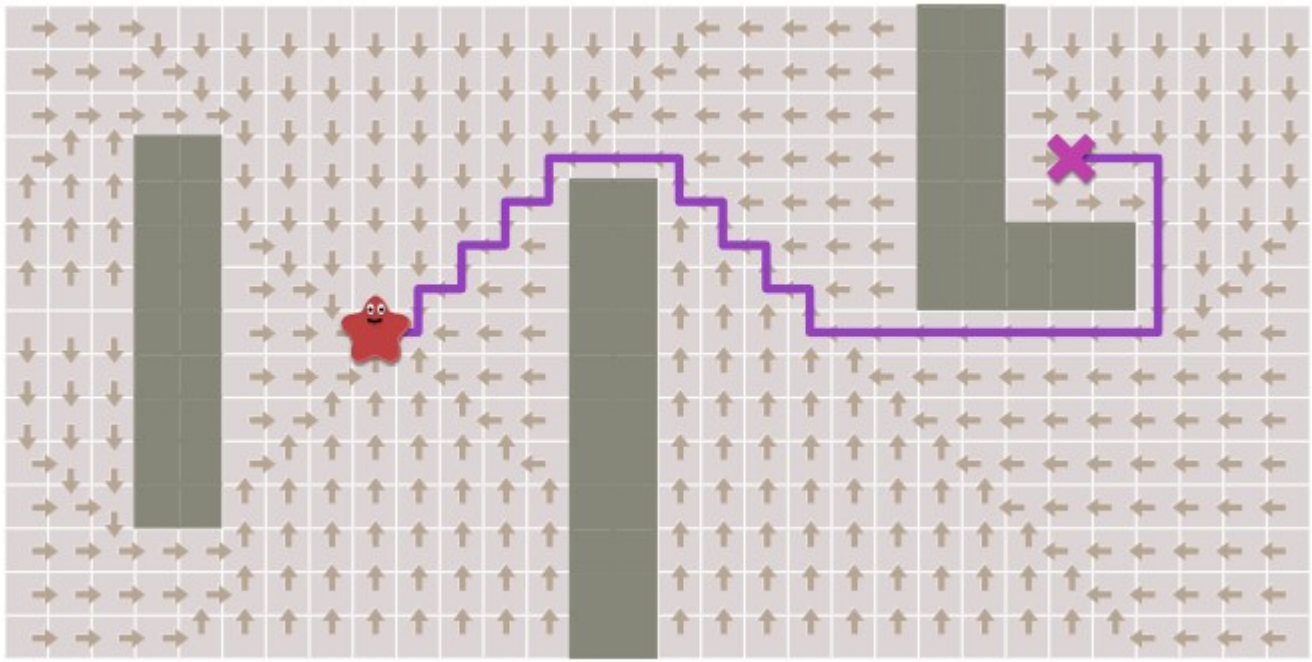
It's only ten lines of (Python) code:

```
frontier = Queue()
frontier.put(start )
reached = set()
reached.add(start)

while not frontier.empty():
   current = frontier.get()
   for next in graph.neighbors(current):
      if next not in reached:
         frontier.put(next)
         reached.add(next)
```



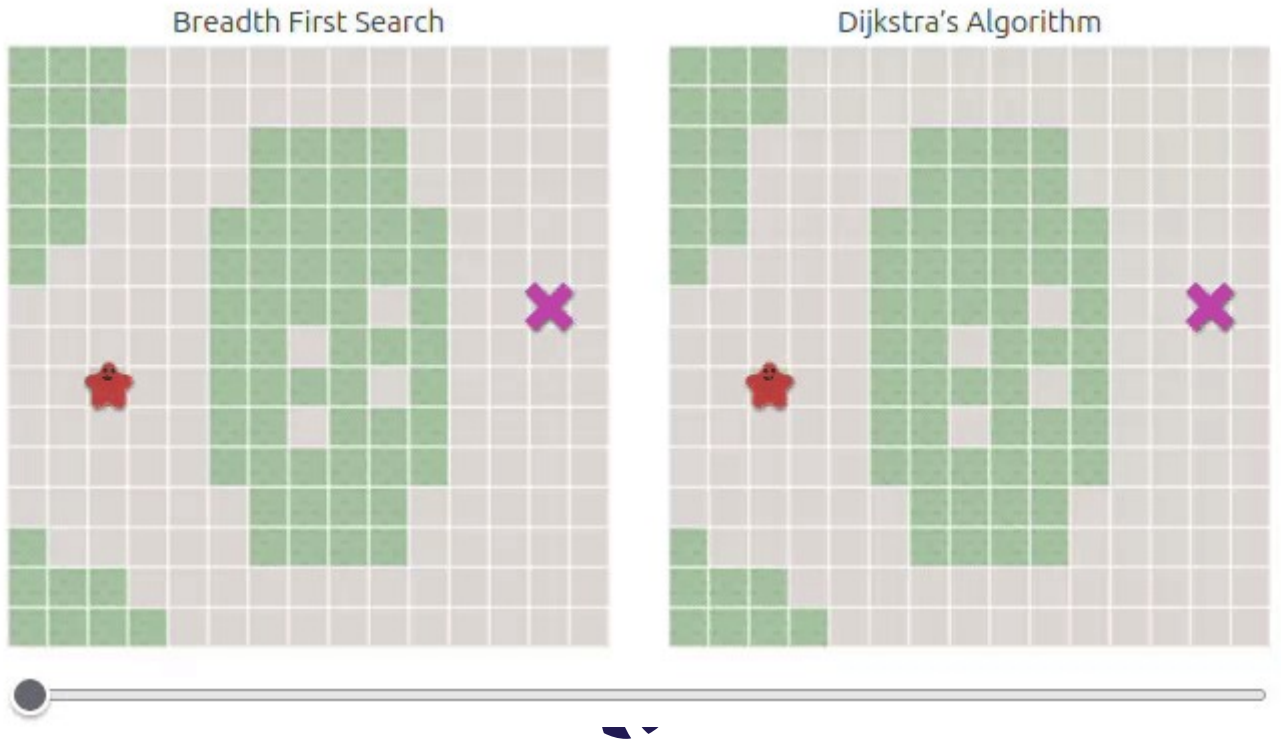
Breadth first search (BFS) – the simplest approach



Movement costs – a step up in complexity

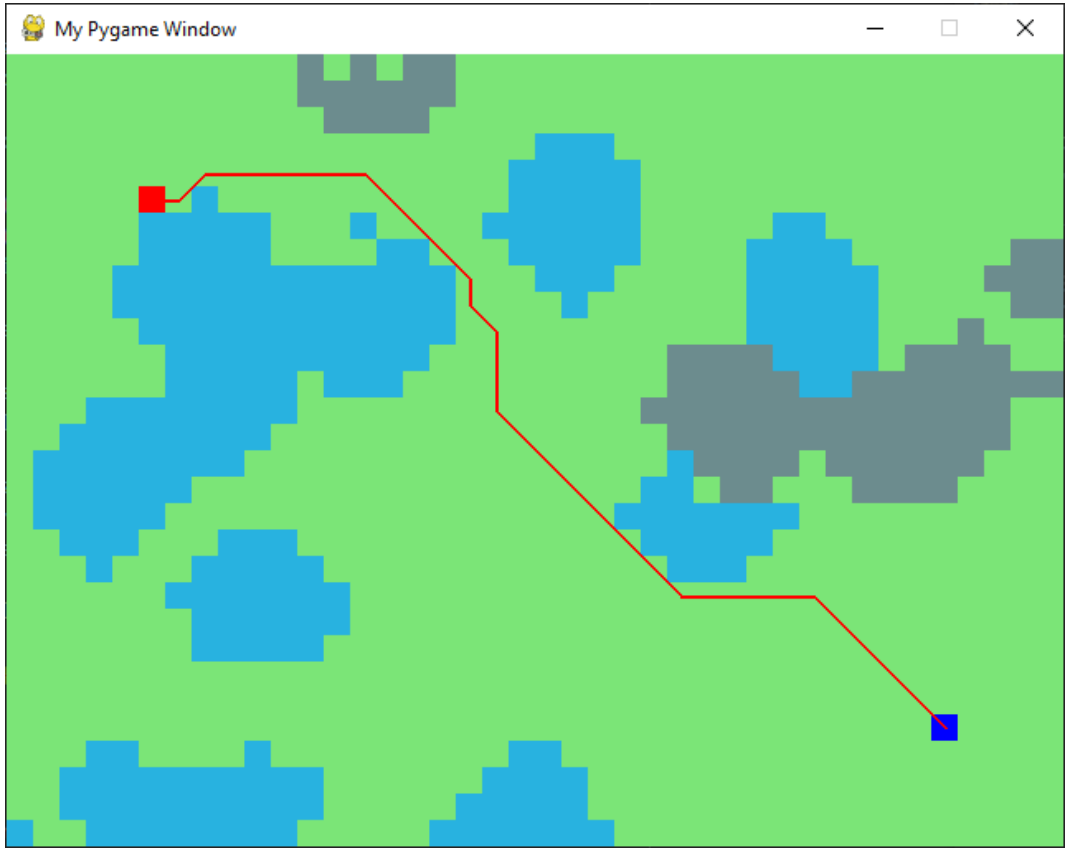


Movement costs – a step up in complexity



The recipe

- Make a map (use lecture 7 exercises)
  - Simpler: Hardcode a map
  - A step up: Autogenerate an “interesting” map
- Make a flood-fill algorithm
- Keep track of direction
  - A step up: Keep track of cost
- Retrace the route
- Done!



Miniprojekt 2

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